

General Relativity Exercise 1

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1 Warmup

1.1 The Metric is symmetric

The metric defines an inner product, which is symmetric by definition:

$$\begin{aligned} g_{ij}x_iy_j &= g_{ji}x_jy_i \\ g &= g^T \end{aligned} \tag{1}$$

1.2 A non-uniform metric

The most simple example of a non-uniform metric is the metric of the 2d Euclidean space in polar coordinates:

$$g^{ij} = \begin{pmatrix} 1 & \\ & r^2 \end{pmatrix} \tag{2}$$

1.3 Invariance of ds^2 under change of coordinates

Let there be two sets of coordinates x^μ and x'^μ , and the metric is defined for the first set of coordinates such that

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu \tag{3}$$

The transformation between the two sets of coordinates is given by the Jacobian:

$$\begin{aligned} J^\mu_\nu &= \frac{\partial x'^\mu}{\partial x^\nu} & (J^{-1})^\mu_\nu &= \frac{\partial x^\mu}{\partial x'^\nu} \\ dx'^\mu &= J^\mu_\nu dx^\nu \\ g'_{\mu\nu} &= (J^{-1})^\sigma_\mu (J^{-1})^\tau_\nu g_{\sigma\tau} \end{aligned} \tag{4}$$

So, using the property of the Jacobian

$$J^\mu_\alpha (J^{-1})^\beta_\mu = \frac{\partial x'^\mu}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x'^\mu} = \frac{\partial x^\beta}{\partial x^\alpha} = \delta^\beta_\alpha \tag{5}$$

The norm in the x'^μ coordinate system, is

$$ds'^2 = g'_{\mu\nu}dx'^\mu dx'^\nu = (J^{-1})^\sigma_\mu (J^{-1})^\tau_\nu g_{\sigma\tau} J^\mu_\alpha dx^\alpha J^\nu_\beta dx^\beta = \delta^\sigma_\alpha \delta^\tau_\beta g_{\sigma\tau} dx^\alpha dx^\beta = g_{\sigma\tau} dx^\sigma dx^\tau = ds^2 \tag{6}$$

1.4 Another example for the usage of a non-measurable variable

A good example for that would come from the ladder operators in QM. In addition to the x and p operators, we create another form of looking at the data, which is unobservable but useful in certain situations.

2 Properties of the Lorentz Transformation

2.1 The Lorentz Group

2.1.1 Definition

A Lorentz transformation Λ is a linear transformation that preserves the norm $ds^2 = c^2 dt^2 - d\vec{x}^2$.

2.1.2 Multiplication

Let Λ_μ^ν and $\Lambda_\mu^{\nu'}$ be two Lorentz transformations. Their product is also linear, since the product of the two linear transformation is linear.

Now, let v^μ be a vector and ds^2 its squared norm. By the definition, the vector $v'^\mu = \Lambda_\nu^\mu v^\nu$ has the same norm ds^2 , and again, by the definition, the vector $v''^\mu = \Lambda_\nu^{\mu'} \Lambda_\sigma^{\nu'} v^\sigma$ has the same norm as v'^μ and v^μ . This means that the product of the two transformations $\Lambda' \circ \Lambda$ also preserves the norm and therefore a Lorentz transformation.

2.1.3 The Lorentz Group

To demonstrate that the set of all Lorentz transformations over a given Minkowski space is a group, we must show that it satisfies the four group axioms.

The first axiom is closure: the product of two items of the group is also an item of the group. We demonstrated that in section 2.1.2.

The second axiom is associativity, which is an inherit part of linear transformation product and therefore needs no additional proof.

The third axiom is the existence of an identity element. It's easy to see that the identity matrix follows the definition of a Lorentz transformation, and therefore is a part of the group.

The last axiom is the existence of an inverse element. Since all Lorentz transformations are linear, we can represent them in a generic base. Because the transformations preserve the Minkowski norm, their determinant must be ± 1 and not zero. This means they must be invertible.

These four conditions are the four axioms of a group, meaning they are enough to prove that the set of all Lorentz Transformations is a group.

2.1.4 Proper and Improper Lorentz Transformations

A Lorentz Transformation is proper if $\det(\Lambda) = 1$ and improper if $\det(\Lambda) = -1$. The product of two proper transformations is also proper, since the determinant of a product of matrices is the product of the determinants:

$$\det(\Lambda_{p1} \circ \Lambda_{p2}) = \det(\Lambda_{p1}) \det(\Lambda_{p2}) = 1 \cdot 1 = 1 \quad (7)$$

In addition, for a given proper Lorentz transformation Λ_{p1} , it is clear from the rule for the determinant of an inverse matrix that Λ_{p1}^{-1} is also proper. This means that the set of all proper Lorentz transformations over a given Minkowski space is closed under multiplication and the inverse operation, and thus is a subset of the Lorentz group.

An example of an improper Lorentz Transformation would be:

$$\Lambda = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \quad (8)$$

This is the time reversal Transformation, and it's a Lorentz Transformation since for an infinitesimal vector $dx^\mu = (dt, d\vec{x})$ the transformed vector is $x'^\mu = \Lambda_\nu^\mu x^\nu = (-dt, d\vec{x})$, and the new norm is $ds'^2 = (-dt)^2 + d\vec{x}^2 = dt^2 + d\vec{x}^2 = ds^2$.

2.1.5 Orthochronous and Antichronous Lorentz Transformation

It is easy to see, using the Lorentz invariance of the infinitesimal line segment, that $|\Lambda_0^0| > 1$:

$$\begin{aligned} ds^2 &= ds'^2 \\ ds^2 &= ds \cdot ds = \eta_{\alpha\beta} ds^\alpha ds^\beta \\ ds'^2 &= ds' \cdot ds' = \Lambda(ds) \cdot \Lambda(ds) = \eta_{\alpha\beta} \Lambda_\gamma^\alpha ds^\gamma \Lambda_\delta^\beta ds^\delta \\ \eta_{\alpha\beta} ds^\alpha ds^\beta &= \eta_{\alpha\beta} \Lambda_\gamma^\alpha ds^\gamma \Lambda_\delta^\beta ds^\delta \end{aligned} \tag{9}$$

This is true for every ds , therefore

$$\eta_{\alpha\beta} = \eta_{\gamma\delta} \Lambda_\alpha^\gamma \Lambda_\beta^\delta \tag{10}$$

Where we replaced the indices on the right to match those on the left. Choosing the element for which $\alpha = \beta = 0$, we get $\eta_{00} = 1$

$$\begin{aligned} 1 &= \eta_{\gamma\delta} \Lambda_0^\gamma \Lambda_0^\delta \\ 1 &= (\Lambda_0^0)^2 - \sum (\Lambda_0^i)^2 \\ \Lambda_0^0 &= \pm \sqrt{1 + \sum (\Lambda_0^i)^2} \\ |\Lambda_0^0| &> 1 \end{aligned} \tag{11}$$

Therefore it's enough to show that for orthochronous transformations, $\Lambda_0^0 > 0$. Using the definition of an orthochronous transformation

$$\Lambda(e_0) \cdot e_0 > 0 \tag{12}$$

This means that for every future directed e_0 , $\Lambda(e_0)$ is also future directed: $\Lambda(e_0)^0 > 0$. Since this is true for every e_0 , it is true in particular to $e_0 = (1, 0, 0, 0)$:

$$\Lambda_\mu^0 e_0^\mu = \Lambda_0^0 > 0 \tag{13}$$

An antichronous transformation, on the other hand, will reverse the time direction. An example for that is the time reversal transformation in equation 8.

2.1.6 The Group of Proper Orthochronous Lorentz Transformation

Given two orthochronous Lorentz transformations Λ_1 and Λ_2 , it is clear that their product is also orthochronous. We can see this by applying the first transformation to a future directed vector e_0 to receive another future directed vector $\Lambda_1(e_0)$, and then apply the second transformation to that vector, getting $\Lambda_2(\Lambda_1(e_0))$. This vector is also future directed, since Λ_2 is orthochronous and it acts on the future directed vector $\Lambda_1(e_0)$. But it can also be seen as the product of both transformations acting on e_0 , which also evidently respects time direction. This means the product $\Lambda_2 \circ \Lambda_1$ is orthochronous as well.

It is also clear that the unit transformation is orthochronous, and that the inverse transformation of an orthochronous transformation is also orthochronous. This is because acting upon the defining equation using the inverse matrix from the left cancels out the original matrix, and leaves it to act on the original vector. In other words, given a orthochronous transformation $\Lambda(e_0) \cdot e_0 > 0$, it is clear that $\Lambda^{-1}(\Lambda(e_0)) \cdot \Lambda(e_0) = e_0 \cdot \Lambda(e_0) > 0$, which means that for the future directed vector $\Lambda(e_0)$ the transformation Λ^{-1} preserves time direction. Since we can choose e_0 however we want, Λ^{-1} is orthochronous.

Putting these properties together we conclude that the set of all proper orthochronous Lorentz transformations is a group.

2.2 On Boosts

2.2.1 Boost in an arbitrary direction

To get a boost in an arbitrary direction, we simply use a rotation matrix with angles θ and φ turning the system around the x^2 and x^3 axes respectively:

$$R_y(\theta) = \begin{pmatrix} 1 & & & \\ & \cos \theta & & -\sin \theta \\ & & 1 & \\ & \sin \theta & & \cos \theta \end{pmatrix} \quad R_z(\varphi) = \begin{pmatrix} 1 & & & \\ & \cos \varphi & & -\sin \varphi \\ & \sin \varphi & & \cos \varphi \\ & & & 1 \end{pmatrix} \quad (14)$$

Using these matrices to rotate the $\vec{\beta} = \beta \hat{x}$ vector and the Λ Lorentz transformation along the x axis, around the y axis and then the z axis, we get:

$$\begin{aligned} \beta' &= R_3(\varphi)R_2(\theta)\vec{\beta} = \beta \begin{pmatrix} \cos \theta \cos \varphi \\ \sin \varphi \\ \sin \theta \cos \varphi \end{pmatrix} \equiv \begin{pmatrix} \beta^1 \\ \beta^2 \\ \beta^3 \end{pmatrix} \\ \Lambda' &= R_2(\theta)R_3(\varphi)\Lambda R_3(-\varphi)R_2(-\theta) = \\ &= \begin{pmatrix} \gamma & -\gamma\beta \cos \theta \cos \varphi & -\gamma\beta \sin \varphi & -\gamma\beta \sin \theta \cos \varphi \\ -\gamma\beta \cos \theta \cos \varphi & 1 + (\gamma - 1) \cos^2 \theta \cos^2 \varphi & (\gamma - 1) \cos \theta \cos \varphi \sin \varphi & (\gamma - 1) \cos^2 \varphi \cos \theta \sin \theta \\ -\gamma\beta \sin \varphi & (\gamma - 1) \cos \theta \cos \varphi \sin \varphi & 1 + (\gamma - 1) \sin^2 \varphi & (\gamma - 1) \sin \theta \cos \varphi \sin \varphi \\ -\gamma\beta \sin \theta \cos \varphi & (\gamma - 1) \cos^2 \varphi \cos \theta \sin \theta & (\gamma - 1) \sin \theta \cos \varphi \sin \varphi & 1 + (\gamma - 1) \sin^2 \theta \cos^2 \varphi \end{pmatrix} \end{aligned} \quad (15)$$

We can replace the angles in the matrix with the respected values of β .

$$\Lambda' = \begin{pmatrix} \gamma & -\gamma\beta^1 & -\gamma\beta^2 & -\gamma\beta^3 \\ -\gamma\beta^1 & 1 + (\gamma - 1) \left(\frac{\beta^1}{\beta}\right)^2 & (\gamma - 1) \frac{\beta^1 \beta^2}{(\beta)^2} & (\gamma - 1) \frac{\beta^1 \beta^3}{(\beta)^2} \\ -\gamma\beta^2 & (\gamma - 1) \frac{\beta^1 \beta^2}{(\beta)^2} & 1 + (\gamma - 1) \left(\frac{\beta^2}{\beta}\right)^2 & (\gamma - 1) \frac{\beta^2 \beta^3}{(\beta)^2} \\ -\gamma\beta^3 & (\gamma - 1) \frac{\beta^1 \beta^3}{(\beta)^2} & (\gamma - 1) \frac{\beta^2 \beta^3}{(\beta)^2} & 1 + (\gamma - 1) \left(\frac{\beta^3}{\beta}\right)^2 \end{pmatrix} \quad (16)$$

And writing the components gives us:

$$\begin{aligned} \hat{x}^0 &= \gamma(x^0 - \vec{\beta} \cdot \vec{x}) \\ \hat{\vec{x}} &= \vec{x} + \frac{\gamma - 1}{\beta^2} (\vec{\beta} \cdot \vec{x}) \vec{\beta} - \gamma \vec{\beta} x^0 \end{aligned} \quad (17)$$

2.2.2 Relativistic velocity addition of velocities

Our new velocity will be $\hat{\vec{u}} = d\hat{\vec{x}}/d\hat{t}$ where $\hat{t} = \hat{x}^0/c$. Using the last section, we can write the infinitesimal length in each coordinate

$$\begin{aligned} d\hat{x}^0 &= \gamma dx^0 - \vec{\beta} \cdot d\vec{x} = \gamma \left(1 - \vec{\beta} \cdot \frac{\vec{u}}{c}\right) dx^0 \\ d\hat{\vec{x}} &= d\vec{x} + \frac{\gamma - 1}{\beta^2} (\vec{\beta} \cdot d\vec{x}) \vec{\beta} - \gamma \vec{\beta} dx^0 = \frac{1}{c} \left(\vec{u} + \frac{\gamma - 1}{\beta^2} (\vec{\beta} \cdot \vec{u}) \vec{\beta} - \gamma c \vec{\beta}\right) dx^0 \end{aligned} \quad (18)$$

Splitting $\vec{u}$ into $\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$, we get

$$\begin{aligned} d\hat{x}^0 &= \gamma \left(1 - \beta \frac{u_{\parallel}}{c}\right) dx^0 \\ d\hat{\vec{x}} &= \frac{1}{c} \left(\vec{u}_{\perp} + \gamma \vec{u}_{\parallel} - \gamma c \vec{\beta}\right) dx^0 \end{aligned} \quad (19)$$

And the velocity is

$$\hat{u} = c \frac{d\hat{x}}{d\hat{x}^0} = \frac{\vec{u}_\perp + \gamma \vec{u}_\parallel - \gamma c \vec{\beta}}{\gamma (1 - \beta \frac{u_\parallel}{c})} \quad (20)$$

And for $c \rightarrow \infty$, which gives $\beta = 0$, $\vec{\beta}c = \vec{v}$, and $\gamma = 1$, the denominator becomes 1 and we get:

$$\hat{u} = \vec{u}_\perp + \vec{u}_\parallel - \vec{v} = \vec{u} - \vec{v} \quad (21)$$

3 Rocket

3.1 Non-Relativistic Rocket

3.1.1 Instantaneously Infinitesimal Change in the Rocket's Velocity in its reference frame

Because the burning rate is constant, the masses of the rocket and the gasses expulted as a function of time is

$$\begin{aligned} M_r(t) &= M_i + \frac{M_f - M_i}{T} t \\ M_g(t) &= \frac{M_i - M_f}{T} t \end{aligned} \quad (22)$$

Since we assume an inertial frame of reference, we can use the preservation of momentum. The infinitesimal momentum leaving the rocket is

$$dp = dM_g(t)v_e + M_g(t)dv_e = \frac{M_i - M_f}{T} v_e dt \quad (23)$$

Because of the preservation of momentum, this is also the infinitesimal increase in the rocket's momentum to the opposite direction:

$$dp = dM_r(t)v_r + M_r(t)dv_r = \frac{M_f - M_i}{T} v_r dt + \left(M_i + \frac{M_f - M_i}{T} t \right) dv_r = \frac{M_i - M_f}{T} v_e dt \quad (24)$$

Assigning $v_r = 0$, since the rocket is at rest, and solving for dv_r , we get:

$$dv_r = \frac{M_i - M_f}{M_i T + (M_f - M_i)t} v_e dt \quad (25)$$

3.1.2 The Acceleration $a(t)$

Simply dividing equation 22 by dt gives

$$a(t) = \frac{M_i - M_f}{M_i T + (M_f - M_i)t} v_e = \frac{\mu}{1 - \mu\tau} \frac{v_e}{T} \quad (26)$$

Where

$$\mu = \frac{M_i - M_f}{M_i} \quad \tau = \frac{t}{T} \quad (27)$$

3.1.3 Final Velocity

By integrating over equation 22 from $t = 0$ to $t = T$ on the right and from $v = 0$ to $v = v_f$ on the left, we get

$$v_f = v_e \int_0^1 \frac{\mu}{1 - \mu\tau} d\tau = -v_e [\ln(1 - \mu\tau)]_0^1 = v_e \ln \frac{1}{1 - \mu} = v_e \ln \frac{M_i}{M_f} \quad (28)$$

It can be seen that since the rocket's mass continues to decrease, the constant expulsion of gas, while increasing its momentum by a constant amount, increases the rocket's velocity by a uniformly increasing amount, causing it to eventually surpass the gas expulsion velocity.

3.2 Relativistic Rocket

3.2.1 Moving between reference frames

When moving between inertial reference frames, the acceleration follows the Lorentz transformation, but also follows the formula in equation 23. For instance, moving to the inertial reference frame where $v_e = 0$ would cause the acceleration to be zero as well, which of course cannot be the case.

3.2.2 Momentum and Energy Conservation in a generic reference frame

In a generic inertial reference frame, where the velocities of the rocket and the gas are v_1 and v_2 respectively, and their masses are m_1 and dm_2 respectively, the conservation of energy and momentum can be formulated as:

$$\begin{aligned} m_1 v_1 &= (m_1 + dm_1)(v_1 + dv_1) + dm_2 v_2 \\ \frac{1}{2} m_1 v_1^2 &= \frac{1}{2} (m_1 + dm_1)(v_1 + dv_1)^2 + \frac{1}{2} dm_2 v_2^2 \end{aligned} \quad (29)$$

Or

$$\begin{aligned} 0 &= m_1 dv_1 + dm_1 v_1 + dm_1 dv_1 + dm_2 v_2 \\ 0 &= 2m_1 v_1 dv_1 + m_1 dv_1^2 + dm_1 v_1^2 + 2dm_1 v_1 dv_1 + dm_1 dv_1^2 + dm_2 v_2^2 \end{aligned} \quad (30)$$

3.2.3 The masses dm_1 and dm_2 as a function of m_1 , v_1 , v_2 , and dv_1

Solving equation 30 and dropping all the quadratic differential terms, we get

$$\begin{aligned} (v_1 + dv_1)dm_1 + v_2 dm_2 &= -m_1 dv_1 \\ (v_1 + dv_1)^2 dm_1 + v_2^2 dm_2 &= -2m_1 v_1 dv_1 + m_1 dv_1^2 \\ v_1 dm_1 + v_2 dm_2 &= -m_1 dv_1 \\ v_1^2 dm_1 + v_2^2 dm_2 &= -2m_1 v_1 dv_1 \end{aligned} \quad (31)$$

$$dm_2 = m_1 \frac{v_1}{v_1 - v_2} \frac{dv_1}{v_2} \quad dm_1 = -m_1 \frac{2v_1 - v_2}{v_1 - v_2} \frac{dv_1}{v_1}$$

3.2.4 v_e as a function of v_1 and v_2

Solving the last part in equation 31 we get

$$\frac{dv_1}{dm_1} = -\frac{1}{m_1} \frac{v_1(v_1 - v_2)}{2v_1 - v_2} \quad (32)$$

We can rename the velocities, where v_1 is the velocity of the rocket in our frame of reference $\hat{v}_r$, and $v_2 - v_1$ is the velocity of the gas relative to the rocket in our frame of reference $\hat{v}_e$. We will also relabel the mass m_1 to $\hat{m}_1$ to mark that it is also the mass in our frame of reference:

$$\frac{d\hat{v}_r}{d\hat{m}_1} = -\frac{1}{\hat{m}_1} \frac{\hat{v}_r \hat{v}_e}{\hat{v}_r - \hat{v}_e} \quad (33)$$

Moving to the rocket's frame of reference, with $\beta = \hat{v}_1/c$, we get:

$$\begin{aligned} v_r &= 0 \\ v_e &= \frac{\beta c + \hat{v}_e}{c + \beta \hat{v}_e} c = \frac{v_2}{c \left(1 + \frac{v_1(v_2 - v_1)}{c^2} \right)} \end{aligned} \quad (34)$$

3.2.5 The mass ratio for a given v_f

Solving the QED in equation 33 by separation we get

$$\ln(\hat{v}_r) + \ln(\hat{v}_e) = -\ln(\hat{m}_1) + \alpha$$

$$\frac{\hat{M}_f}{\hat{M}_i} = \frac{\hat{v}_{r,f}\hat{v}_{e,f}}{\hat{v}_{r,i}\hat{v}_{e,i}} \quad (35)$$

Using equation 20 to derive the opposite transformation to move to the rocket's frame of reference, the mass ratio remains the same and we get:

$$\hat{v}_r = \beta c \quad \hat{v}_e = \frac{v_e - \beta c}{c - \beta v_e} c$$

$$\frac{M_f}{M_i} = \frac{\beta_f}{\beta_i} \cdot \frac{v_e - \beta_f c}{v_e - \beta_i c} \cdot \frac{c - \beta_i v_e}{c - \beta_f v_e} \quad (36)$$

Marking $v_f = \beta_f c$ and $v_i = \beta_i c$ we get

$$\frac{M_f}{M_i} = \frac{v_f}{v_i} \cdot \frac{v_e - v_f}{v_e - v_i} \cdot \frac{c^2 - v_i v_e}{c^2 - v_f v_e} \quad (37)$$

It can be seen that when $v_e \gg v_i, v_f$, the first fraction on the right remains the same, the second fraction goes to one, and the third fraction is opposite to the first fraction, leaving the entire right side to go to one. This means that for a very fast fuel, the fraction of rocket's mass used for fuel can be very small. Equation 28 shows the result in the Newtonian limit, where dividing by v_e to get a unitless equation gives

$$\frac{v_f}{v_e} = \ln \frac{M_i}{M_f} \quad (38)$$

And taking v_e to be very large result in the same situation when $\frac{M_i}{M_f} \rightarrow 1$.